Assignment 4

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T-519-STOR: Theory of Computation

Problem 1: the complexity class NP (30 points)

1. (15 points) Assume $A \in$ NP and $B \in$ P. Show that $A \setminus B \in$ NP

Solution:
Let M_A be a nondeterministic Turing machine that decides A in polynomial time, Let M_B be a deterministic Turing machine that decides B in polynomial time, we construct a nondeterministic Turing machine M that decides $A \setminus B$ in polynomial time:

Let $M =$ "On input $\langle w \rangle$:

1. Run M_A on input $\langle w \rangle$
2. If M_A rejects, reject,otherwise run M_B on w
3. If M_B accepts, reject, if M_B rejects, accept"

Correctness:

As $A \setminus B$ means the set of strings that are in A but not in B, the decider M runs M_A to check if w is in A, if it is in A, we need to check if w is in B, as if it is in B, then it cannot be in$A \setminus B$, thus we accept only if w is in A but not in B, for time, since we know A is in NP, we know that M_A runs in polynomial time for a nondeterministic Turing machine, so running M_A takes polynomial time for M as M is nondeterministic, similarly M_B takes polynomial time for a deterministic Turing machine, so it takes polynomial time for a nondeterministic Turing machine, thus M runs in polynomial time and as it decides $A \setminus B$, $A \setminus B \in$ NP

2. (15 points) Assume that P = NP. Give a polynomial-time algorithm for finding a satisfying assignment for a boolean formula ϕ, if one exists.

Solution:

Problem 2: Polynomial time reductions and NP-completeness

1. (20 points) Consider the language

$$B = \{\langle G, k, v\rangle \mid G \text{ is a graph with a } k\text{-clique containing } v\}$$

(a) Show that B is in NP.

(b) Give a polynomial-time reduction from CLIQUE to B.

(c) Is B NP-complete?

Solution:

Recall that for a graph, we denote it with $G = (V, E)$, where V is the set of vertices, let M_C be a nondeterministic Turing machine that decides CLIQUE that runs in polynomial time, we construct a nondeterministic Turing machine M that decides B.

Let $M =$ "On input $\langle G, k, v\rangle$:

1. For each $w \in V$, we check if $w = v$
2. If there exists a w, such that $w = v$, we run M_C on $\langle G, k\rangle$ and goto step 3, if no such w exists, we reject.
3. If M_C accepts, accepts, otherwise reject"

Correctness:

Based on the definition of B, it is almost the same as CLIQUE except the added condition of containing v, so, we start by checking if $v \in V$ and if so, we can just run the decider for CLIQUE, if it doesn't exist, then G does not contain v and we reject. CLIQUE is in NP, so there exists a nondeterministic Turing machine that decides CLIQUE in polynomial time, and as we have one for loop that goes through V, its also polynomial, so this machine runs in polynomial time and as such $B \in$ NP